

# Package ‘rENM.core’

June 21, 2026

**Type** Package  
**Title** Shared utilities for the rENM Framework  
**Version** 0.1.0  
**Description** Provides shared infrastructure for the rENM Framework, including project directory resolution, species and environmental variable metadata lookup, and logging utilities used across all framework packages.  
**Depends** R (>= 4.1.0)  
**License** MIT + file LICENSE  
**Encoding** UTF-8  
**Roxygen** list(markdown = TRUE)  
**URL** <https://github.com/rENM-Framework/rENM.core>  
**BugReports** <https://github.com/rENM-Framework/rENM.core/issues>  
**Imports** utils  
**Config/roxygen2/version** 8.0.0

## Contents

|                            |          |
|----------------------------|----------|
| get_species_info . . . . . | 1        |
| rENM_project_dir . . . . . | 2        |
| show_species . . . . .     | 4        |
| show_variables . . . . .   | 4        |
| <b>Index</b>               | <b>6</b> |

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|------------------|-----------------------------------------------------------|
| get_species_info | <i>Look up species metadata by four-letter alpha code</i> |
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## Description

Reads the project species table (data/\_species.csv) and returns a one-row data frame of standardized metadata for the requested species. Matching is case-insensitive. Column names in the source file are normalized before lookup, so minor variations (e.g., ALPHA.CODE vs SPECIESCODE) are handled automatically.

**Usage**

```
get_species_info(alpha_code, project_dir = NULL)
```

**Arguments**

|             |                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alpha_code  | Character scalar. Four-letter species alpha code (e.g., "CASP"). Case-insensitive; leading/trailing whitespace is trimmed.                                                             |
| project_dir | Character. Path to the rENM project root. If NULL (default), resolved via <a href="#">rENM_project_dir</a> (argument, rENM.project_dir option, RENM_PROJECT_DIR environment variable). |

**Details****Alpha-code column detection**

The function searches for the alpha-code column by normalizing all column names to uppercase with punctuation removed, then matching against these candidates in order: ALPHACODE, ALPHA, SPECIESCODE, SPSC, BANDINGCODE. The first match wins. If none is found, the function errors with the list of available columns.

**Output column standardization**

Regardless of source column names, the returned data frame always uses these names: COMMON.NAME, SCIENTIFIC.NAME, EBD.RECORDS, EBD.RANGE, GAP.RANGE. The same normalization logic is used to locate these columns in the source file.

**Value**

A one-row data frame with columns COMMON.NAME, SCIENTIFIC.NAME, EBD.RECORDS, EBD.RANGE, and GAP.RANGE. Row names are reset to NULL.

**See Also**

[rENM\\_project\\_dir](#), [show\\_species](#)

**Examples**

```
## Not run:
info <- get_species_info("CASP")
info <- get_species_info("CASP", project_dir = "/projects/rENM")

## End(Not run)
```

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rENM\_project\_dir

*Resolve and validate the rENM project root directory*


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**Description**

Returns a normalized absolute path to the rENM project root, resolving it from the first available source in this precedence order: the explicit project\_dir argument, the rENM.project\_dir R option, or the RENM\_PROJECT\_DIR environment variable. Errors with an actionable message if no valid directory can be found.

**Usage**

```
rENM_project_dir(project_dir = NULL)
```

**Arguments**

**project\_dir**      Character. Path to the rENM project root. Takes precedence over the R option and environment variable. If NULL (default), the option and environment variable are tried in order.

**Details**

This function is the single point of filesystem configuration for the rENM framework. All other functions that access the project root call this one rather than resolving paths independently, keeping hard-coded paths out of the codebase and making scripts portable and CRAN-compliant.

A candidate path is accepted only if it is a non-empty scalar character string that expands without error (`path.expand`) and refers to a directory that already exists (`normalizePath(..., mustWork = TRUE)`). A candidate that fails any check is silently skipped and the next source is tried.

**Configuring the project directory**

**Explicit argument (recommended for scripts and tests)** Pass the path directly: `rENM_project_dir("/projects/rENM")`

**R option (session-level convenience)** `options(rENM.project_dir = "~/rENM")`

**Environment variable (persistent across sessions)** Add `RENM_PROJECT_DIR=~/rENM` to `~/.Renviron`, or call `Sys.setenv(RENM_PROJECT_DIR = "~/rENM")` for the current session.

**Value**

Character scalar. Normalized absolute path to an existing directory.

**See Also**

[get\\_species\\_info](#), [show\\_species](#), [show\\_variables](#)

**Examples**

```
## Not run:
# Explicit path – best for reproducible scripts and tests
rENM_project_dir("/projects/rENM")

# Session option
options(rENM.project_dir = "~/rENM")
rENM_project_dir()

# Environment variable
Sys.setenv(RENM_PROJECT_DIR = "~/rENM")
rENM_project_dir()

## End(Not run)
```

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|              |                                                       |
|--------------|-------------------------------------------------------|
| show_species | <i>Print the project species table to the console</i> |
|--------------|-------------------------------------------------------|

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### Description

Reads data/\_species.csv from the rENM project directory and prints its contents with a timestamped header. Intended as a lightweight inspection utility for verifying species definitions before running the pipeline.

### Usage

```
show_species(project_dir = NULL)
```

### Arguments

|             |                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| project_dir | Character. Path to the rENM project root. If NULL (default), resolved via <a href="#">rENM_project_dir</a> (argument, rENM.project_dir option, RENM_PROJECT_DIR environment variable). |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Value

Invisibly returns the species data frame. Called primarily for the side effect of printing a formatted table and header to the console.

### See Also

[rENM\\_project\\_dir](#), [get\\_species\\_info](#), [show\\_variables](#)

### Examples

```
## Not run:
show_species()
show_species(project_dir = "/projects/rENM")

## End(Not run)
```

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|                |                                                         |
|----------------|---------------------------------------------------------|
| show_variables | <i>Print the project variables table to the console</i> |
|----------------|---------------------------------------------------------|

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### Description

Reads data/\_variables.csv from the rENM project directory and prints its contents with a timestamped header. Intended as a lightweight inspection utility for verifying environmental variable definitions before running the pipeline.

### Usage

```
show_variables(project_dir = NULL)
```

**Arguments**

`project_dir`      Character. Path to the rENM project root. If NULL (default), resolved via [rENM\\_project\\_dir](#) (argument, `rENM.project_dir` option, `RENM_PROJECT_DIR` environment variable).

**Value**

Invisibly returns the variables data frame. Called primarily for the side effect of printing a formatted table and header to the console.

**See Also**

[rENM\\_project\\_dir](#), [show\\_species](#)

**Examples**

```
## Not run:  
show_variables()  
show_variables(project_dir = "/projects/rENM")  
  
## End(Not run)
```

# Index

`get_species_info`, [1](#), [3](#), [4](#)

`rENM_project_dir`, [2](#), [2](#), [4](#), [5](#)

`show_species`, [2](#), [3](#), [4](#), [5](#)

`show_variables`, [3](#), [4](#), [4](#)